

Q1 - 24 January - Shift 2

A parallel plate capacitor with air between the plate has a capacitance of 15pF. The separation between the plate becomes twice and the space between them is filled with a medium of dielectric constant 3.5. Then the capacitance becomes $\frac{x}{4}$ pF.

The value of x is .

Space for your notes:

Q2 - 25 January - Shift 1

A parallel plate capacitor has plate area 40cm^2 and plates separation 2 mm. The space between the plates is filled with a dielectric medium of a thickness 1 mm and dielectric constant 5. The capacitance of the system is :

Space for your notes:

- (1) $24\varepsilon_0 \text{ F}$ (2) $\frac{3}{10}\varepsilon_0 \text{ F}$
(3) $\frac{10}{3}\varepsilon_0 \text{ F}$ (4) $10\varepsilon_0 \text{ F}$

Q3 - 25 January - Shift 2

A capacitor has capacitance $5\mu\text{F}$ when it's parallel plates are separated by air medium of thickness d.

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A slab of material of dielectric constant 1.5 having area equal to that of plates but thickness $\frac{d}{2}$ is inserted between the plates. Capacitance of the capacitor in the presence of slab will be $\underline{\hspace{1cm}}\mu\text{F}$.

Q4 - 30 January - Shift 1

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A capacitor of capacitance $900\ \mu\text{F}$ is charged by a $100\ \text{V}$ battery. The capacitor is disconnected from the battery and connected to another uncharged identical capacitor such that one plate of uncharged capacitor connected to positive plate and another plate of uncharged capacitor connected to negative plate of the charged capacitor. The loss of energy in this process is measured as $x \times 10^{-2}\ \text{J}$. The value of x is .

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Q5 - 31 January - Shift 2

Two parallel plate capacitors C_1 and C_2 each having capacitance of $10\ \mu\text{F}$ are individually charged by a $100\ \text{V}$ D.C. source. Capacitor C_1 is kept connected to the source and a dielectric slab is inserted between its plates. Capacitor C_2 is disconnected from the source and then a dielectric slab is inserted in it. Afterwards the capacitor C_1 is also disconnected from the source and the two capacitors are finally connected in parallel combination. The common potential of the combination will be _____ V. (Assuming Dielectric constant = 10)

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Q6 - 01 February - Shift 1

A charge particle of $2\ \mu\text{C}$ accelerated by a potential difference of $100\ \text{V}$ enters a region of uniform magnetic field of magnitude $4\ \text{mT}$ at right angle to the direction of field. The charge particle completes semicircle of radius $3\ \text{cm}$ inside magnetic field. The mass of the charge particle is _____ $\times 10^{-18}\ \text{kg}$.

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Q7 - 01 February - Shift 2

Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Two metallic spheres are charged to the same potential. One of them is hollow and another is solid, and both have the same radii. Solid sphere will have lower charge than the hollow one.

Reason R : Capacitance of metallic spheres depend on the radii of spheres.

In the light of the above statements, choose the correct answer from the options given below.

- (1) **A** is false but **R** is true
- (2) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) Both **A** and **R** are true but **R** is not the correct explanation of **A**

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Answer Key

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(As per Official NTA Key released on 2 Feb)

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Q1 (105)

Q2 (3)

Q3 (6)

Q4 (225)

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Q5 (55)

Q6 (144)

Q7 (1)

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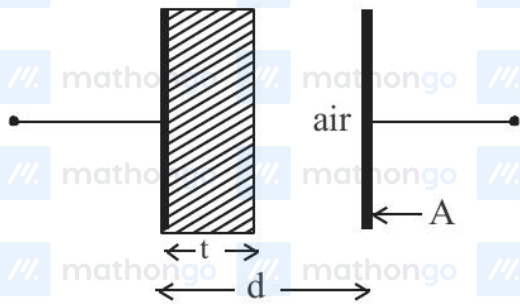
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Q1 (105)

$$C_0 = \frac{\epsilon_0 A}{d} = 15 \text{ pF}$$

$$C = \frac{K \epsilon_0 A}{2d} = \frac{3.5}{2} \times 15 \text{ pF} = \frac{105}{4} \text{ pF}$$

Q2 (3)



This can be seen as two capacitors in series combination so

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{1}{\frac{K \epsilon_0 A}{t}} + \frac{1}{\frac{\epsilon_0 A}{d-t}}$$

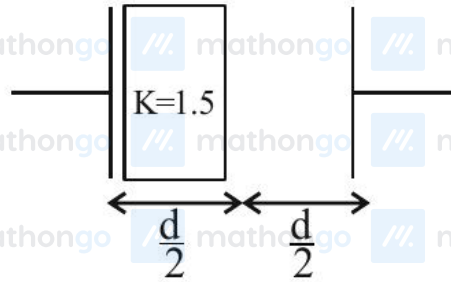
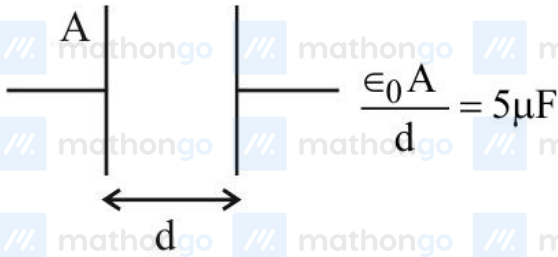
$$= \frac{t}{K \epsilon_0 A} + \frac{d-t}{\epsilon_0 A}$$

$$= \frac{1 \times 10^{-3}}{5 \epsilon_0 \times 40 \times 10^{-4}} + \frac{1 \times 10^{-3}}{\epsilon_0 \times 40 \times 10^{-4}}$$

$$\frac{1}{C_{eq}} = \frac{1}{20 \epsilon_0} + \frac{1}{4 \epsilon_0}$$

$$C_{eq} = \frac{20 \times 4 \epsilon_0}{24} = \frac{10 \epsilon_0}{3} \text{ F}$$

Q3 (6)

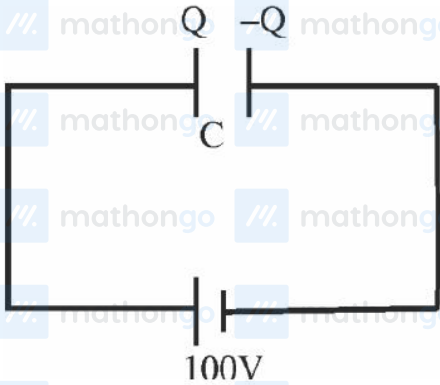


$$C_{\text{new}} = \frac{\epsilon_0 A}{\left(\frac{d}{2}\right) + \left(\frac{d}{2}\right)} = \frac{\epsilon_0 A}{d}$$

$$= \frac{\epsilon_0 A}{\left(\frac{d}{3} + \frac{d}{2}\right)} = \frac{6\epsilon_0 A}{5d}$$

$$= \frac{6}{5} \times 5\mu F = 6\mu F$$

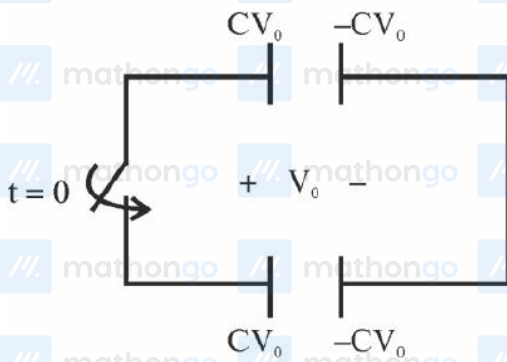
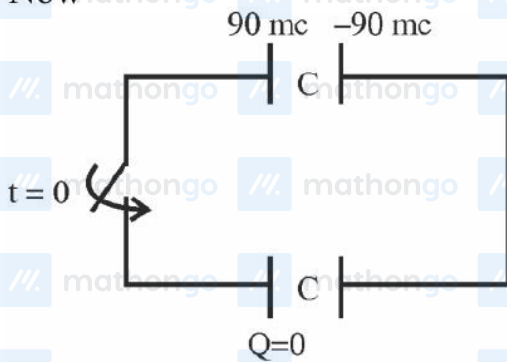
Q4 (225)



$$C = 900 \mu\text{F}$$

$$Q = CV = 900 \times 10^{-6} \times 100 = 9 \times 10^{-2} = 90 \text{ MC}$$

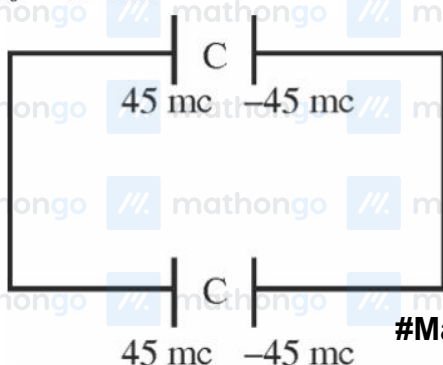
Now



Common potential will be developed across both capacitors by kVL
Total charge on left plates of capacitors should be conserved.

$$90 \text{ mc} + 0 = 2cv_0$$

$$cv_0 = 45 \text{ mc}$$



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To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore
Heat dissipated = $U_i - U_f$ Change in energy

Q5 (55) $\frac{1}{2} \frac{(90 \text{ mc})^2}{900 \mu\text{F}} - 2 \times \frac{1}{2} \frac{(45 \text{ mc})^2}{900 \mu\text{F}} \left[U = \frac{Q^2}{2C} \right]$

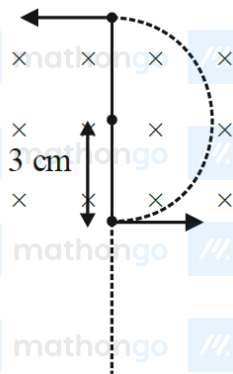
No Solution

= 2.25 Joule

Q6 (144)

OR

$$r = \frac{mv}{qB} = \frac{\sqrt{2km}}{qB}, m = \frac{r^2 q^2 B^2}{2k}$$



$$m = \frac{\frac{1}{100} \times \frac{3}{100} \times 2 \times 2 \times 4 \times 10^{-3} \times 4 \times 10^{-3} \times 10^{-12}}{2 \times (100) \times 10^{-6}} = 144 \times 10^{-18} \text{ kg}$$

Q7 (1)

Potential of a conducting sphere is

$$V = \frac{KQ}{R} \text{ (Solid as well as hollow)}$$

$$V_1 = V_2 \text{ and } R_1 = R_2$$

$$\therefore Q_1 = Q_2$$